

MD. SAKIB AL HASAN

Junior Data Scientist

Narsingdi Sadar, Narsingdi, Bangladesh

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CAREER OBJECTIVE

Junior Data Scientist with hands-on experience in Python, SQL, data analysis, machine learning, and data visualization. Built end-to-end projects involving customer segmentation, classification, EDA, feature engineering, and model evaluation. Seeking an entry-level Data Scientist or Data Analyst role.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, XGBoost), SQL

Data Analysis: Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis

Machine Learning: Regression, Classification, Clustering, Feature Engineering, PCA, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Data Visualization: Matplotlib, Seaborn

Tools: Jupyter Notebook, Git, GitHub, AI-assisted development tools

Additional: Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Microsoft Access

PROJECTS

1) Customer Segmentation using K-Means, Hierarchical & DBSCAN Clustering

[GitHub Repo](#)

- Segmented **8,950 credit card customers** by spending and repayment behavior, engineering **15 features** via log-transforms, correlation filtering, and PCA (9 components, 95% variance).
- Benchmarked **K-Means, Hierarchical, and DBSCAN** ($k=2-10$) via silhouette score; selected **K-Means ($k=2$, silhouette 0.26)** as the best-performing, fully reproducible model in Python (pandas, scikit-learn).
- Identified **two actionable segments** — **60% purchase-active customers vs. 40% cash-advance/high-balance** carriers — and translated each into concrete **marketing and credit-risk recommendations**.

2) Telco Customer Churn Prediction | Python, scikit-learn, XGBoost, SMOTE

[GitHub Repo](#)

- Analyzed **7,043 telecom customers** and compared 6 ML models to predict churn, achieving **0.844 ROC-AUC** and **77% recall** on customers likely to leave.
- Found that **88.6% of churners** are on month-to-month contracts and recommended retention actions (auto-pay incentives, 1-year contract offers) for at-risk customers.

3) Medical Insurance Cost Prediction | Python, scikit-learn, XGBoost, SHAP

[GitHub Repo](#)

- Built an end-to-end regression pipeline on **1,337 insurance records**; engineered smoker $\times$ BMI features that cut baseline cross-validated **RMSE by 25%**, then compared 8 models with 5-fold CV and hyperparameter tuning (XGBoost selected).
- Reached **R^2 of 0.85 (RMSE $\approx$ \$4.6K)** averaged over 20 random splits; SHAP analysis showed **smoking status and age** drive most of the cost, with smokers averaging **$\sim 3.8\times$ higher charges**.

EDUCATION

B.Sc. (Hons) in Mathematics — 4th Year (Running)

Govt. Shaheed Asad College, Shibpur, Narsingdi | Session: 2021–2022 | Expected Completion: 2027

Higher Secondary Certificate (HSC) – Science

Narsingdi Public College, Narsingdi | 2021 | GPA 4.42/5.00

CERTIFICATIONS & TRAINING

- Data Science & Machine Learning with Python** — Ostad 2026 | [View Certificate](#)
- Computer Basic and ICT Applications** — Dept. of Youth Development, Ministry of Youth and Sports, Govt. of Bangladesh (Certificate ID: 141231, Grade: A+, Jul–Dec 2024); intermediate proficiency in MS Word, Excel, Access, and PowerPoint.